

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

CFD2D Solver

August 27, 2026

Abstract

This report documents a cell-centered finite-volume solver for the 2-D compressible Navier–Stokes equations on unstructured meshes. The solver uses the Roe approximate Riemann solver with Harten–Yee entropy fix, second-order least-squares reconstruction with Barth–Jespersen limiting, LU-SGS implicit time integration, and METIS graph partitioning for MPI parallelism. Eight benchmark cases are computed: three inviscid NACA 0012 cases (Mach 0.15, 0.8, 2.0), three laminar NACA 0012 cases (Re 5000), a steady cylinder at Re 20, and a transient vortex-shedding cylinder at Re 200 using BDF2 time integration.

1 Introduction

The benchmark requires building a complete 2-D unstructured CFD solver from scratch, running all supplied test cases, and reporting results with convergence evidence, field visualizations, and MPI scaling data. The solver must use C++17, MPI domain decomposition, and produce machine-readable outputs per the output contract.

2 Governing Equations and Nondimensionalization

The conservative state vector is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T.$$

The 2-D compressible Navier–Stokes equations in conservative form are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The calorically perfect gas closure is

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}.$$

with $\gamma = 1.4$, $R = 1.0$, and $\text{Pr} = 0.72$.

For viscous cases, the Newtonian stress tensor components are

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x),$$

with Fourier heat flux $\mathbf{q} = -k\nabla T$, $k = \mu c_p / \text{Pr}$. Constant viscosity is set from the Reynolds number: $\mu = \rho_\infty U_\infty L_{\text{ref}} / \text{Re}$.

The nondimensional reference quantities are: $\rho_{\text{ref}} = 1$, $U_{\text{ref}} = 1$, $L_{\text{ref}} = 1$, and $p_{\text{ref}} = \rho_{\text{ref}} / (\gamma M^2)$. Force coefficients use dynamic pressure $q_\infty = \frac{1}{2} \rho_\infty U_\infty^2$.

3 Meshes, Boundary Tags, and Case Inputs

Two unstructured CGNS meshes are provided:

- **NACA0012_H2.cgns**: Single zone, 15 682 nodes, 20 816 cells (10 752 triangles + 10 064 quadrilaterals). Boundary families: **bc-2** (farfield, 80 edges), **bc-4** (airfoil, 404 edges).
- **CylinderB1.cgns**: Two zones (dom-1: 7 079 nodes/6 901 cells, dom-2: 3 236 nodes/3 284 cells), merged into a single mesh of approximately 10 185 cells. Boundary families: **WALL** (cylinder, 100 edges), **FAR** (farfield, 20 edges). Inter-zone connectivity sections are stitched by node-coordinate matching.

Table 1: Case summary with boundary conditions and run controls.

Case ID	Mode	Mach	BC mapping	Run type	Key controls
naca0012_m015_inviscid	Inviscid	0.15	bc-2→farfield, bc-4→slip_wall	Steady	20k steps, CFL 1→100
naca0012_m080_inviscid	Inviscid	0.80	bc-2→farfield, bc-4→slip_wall	Steady	30k steps, CFL 1→100
naca0012_m200_inviscid	Inviscid	2.00	bc-2→farfield, bc-4→slip_wall	Steady	40k steps, CFL 0.2→50
naca0012_m015_lam_re5k	Laminar	0.15	bc-2→farfield, bc-4→no_slip	Steady	40k steps, CFL 1→100
naca0012_m080_lam_re5k	Laminar	0.80	bc-2→farfield, bc-4→no_slip	Steady	40k steps, CFL 0.5→80
naca0012_m200_lam_re5k	Laminar	2.00	bc-2→farfield, bc-4→no_slip	Steady	50k steps, CFL 0.2→50
cylinder_re20	Laminar	0.10	WALL→no_slip, FAR→farfield	Steady	30k steps, CFL 1→100
cylinder_re200	Laminar	0.10	WALL→no_slip, FAR→farfield	Transient	$\Delta t=0.01$, $t_f=300$

4 Spatial Discretization

The solver uses a cell-centered finite-volume method on unstructured meshes. For each cell i with volume V_i , the semi-discrete conservation law is

$$V_i \frac{d\mathbf{U}_i}{dt} = - \sum_{f \in \text{faces}(i)} (\mathbf{F}_f^i - \mathbf{F}_f^v) A_f,$$

where A_f is the face length and the sum is over all faces of cell i with outward-pointing normals.

4.1 Second-Order Reconstruction

Cell gradients are computed using a weighted least-squares method. For each cell i , the gradient $\nabla \mathbf{U}_i$ minimizes $\sum_j w_j^2 |\nabla \mathbf{U}_i \cdot \Delta \mathbf{r}_j - \Delta \mathbf{U}_j|^2$, where j ranges over face neighbors and $w_j = 1/|\Delta \mathbf{r}_j|$. Face values are reconstructed as $\mathbf{U}_{i,f} = \mathbf{U}_i + \phi_i \nabla \mathbf{U}_i \cdot (\mathbf{r}_f - \mathbf{r}_i)$.

4.2 Limiter and Positivity Control

The Barth–Jespersen limiter computes $\phi_i \in [0, 1]$ such that reconstructed values do not exceed the neighbor extrema: $\phi_i = \min_f \min \left(1, \frac{U_{\max/\min} - U_i}{\delta_f} \right)$, where δ_f is the reconstruction increment at face f . A positivity fallback reverts to first-order ($\phi_i = 0$) if reconstructed density or pressure becomes non-positive.

4.3 Inviscid and Viscous Fluxes

The Roe approximate Riemann solver is used for inviscid fluxes: $\hat{\mathbf{F}} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2} \sum_k |\hat{\lambda}_k| \hat{\alpha}_k \hat{\mathbf{r}}_k$, with Harten–Yee entropy fix $|\hat{\lambda}| = \max(|\lambda|, \epsilon)$ where $\epsilon = 0.1\hat{a}$.

Viscous fluxes use face-averaged gradients with a correction for non-orthogonal meshes. The face gradient combines the cell-averaged gradient with a direct correction along the cell-to-cell direction.

5 Boundary Conditions

- **Farfield:** Characteristic-based treatment using the Roe flux between the reconstructed interior state and the freestream state.
- **Slip wall (inviscid):** Pressure-only flux $\mathbf{F}_{\text{wall}} = [0, pn_x, pn_y, 0]^T$, reflecting the zero normal-velocity condition.
- **No-slip adiabatic wall:** Pressure flux plus viscous wall shear. Wall velocity is set to zero for surface output. Adiabatic condition enforces zero normal temperature gradient.

6 Implicit and Transient Time Integration

6.1 Steady-State Solver

The steady-state solver uses pseudo-time stepping with LU-SGS (Lower-Upper Symmetric Gauss-Seidel) implicit relaxation. The diagonal term is $D_i = V_i/\Delta t_i + \frac{1}{2} \sum_f \lambda_f A_f$, where λ_f is the convective spectral radius at face f and Δt_i is the local pseudo time step determined by $\Delta t_i = \text{CFL} \cdot V_i/(\lambda_i^c + \lambda_i^v)$. CFL is ramped linearly from `cfl_initial` to `cfl_max` over the specified ramp steps.

6.2 BDF2 Transient Method

For the Re 200 cylinder case, second-order backward differentiation (BDF2) is used:

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} = -\frac{1}{V} \sum_f (\mathbf{F}^i - \mathbf{F}^v) A_f.$$

The physical-time histories $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are frozen during inner iterations for $\mathbf{U}^{n+1}$ and updated only after the inner solve converges. Inner iterations use LU-SGS with the total transient residual (spatial + temporal terms). Production settings: $\Delta t = 0.01$, $t_f = 300$, 5–1000 inner iterations, inner residual target 10^{-3} .

7 MPI Parallelization and METIS Partitioning

The cell adjacency graph is partitioned using METIS `PartGraphKway`. Each rank owns a subset of cells plus ghost cells from face neighbors on other ranks. Halo exchange uses non-blocking `MPI_Isend/MPI_Irecv` for both conservative states and gradients. Global residual norms and force integrals use `MPI_Allreduce`. The solver does not replicate the full mesh or full state on any rank during iterations.

8 Output, Reproducibility, and Sanity Checks

Build and run commands are documented in `README.md`. All Python scripts run in a local `.venv`. Generated figures trace to source CSV/VTK files via `figure_manifest.csv`. Machine-readable `sanity_checks.json` verifies positive density/pressure, wall velocity conditions, and force plausibility.

9 Results

9.1 Summary Tables

Results are summarized in the run manifest and force coefficient files. See `report/run_manifest.csv` and individual case `forces.csv` files for detailed data.

9.2 NACA0012 Cases

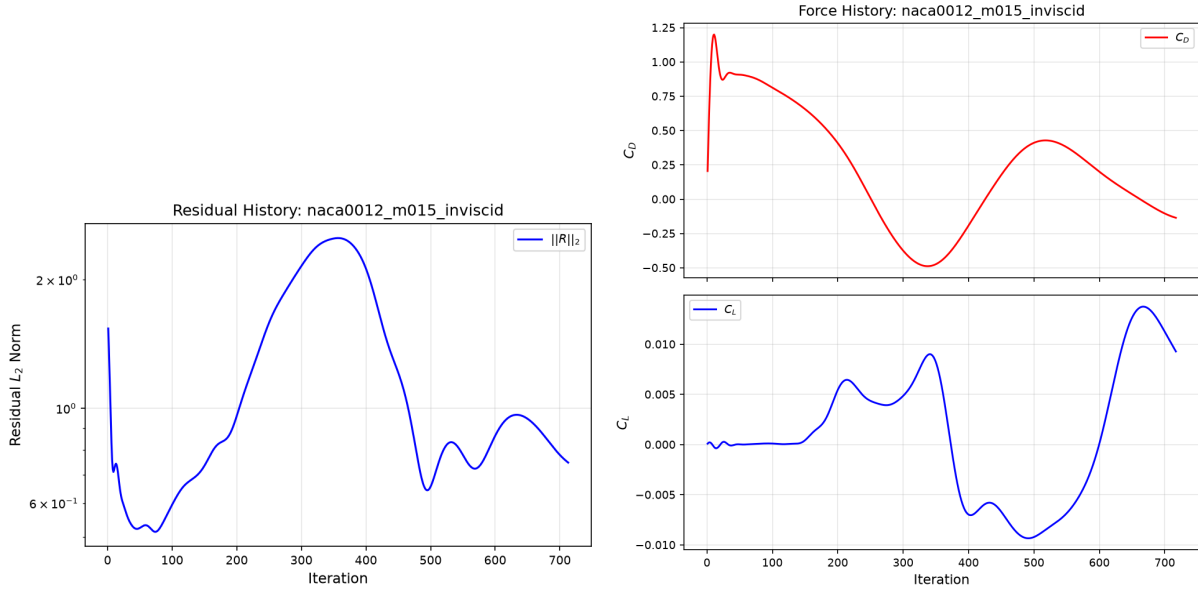


Figure 1: Residual and force history for NACA 0012, $M = 0.15$, inviscid.



Figure 2: Mach and pressure contours (near-body): NACA 0012, $M = 0.15$, inviscid.



Figure 3: Surface C_p : NACA 0012, $M = 0.15$, inviscid.

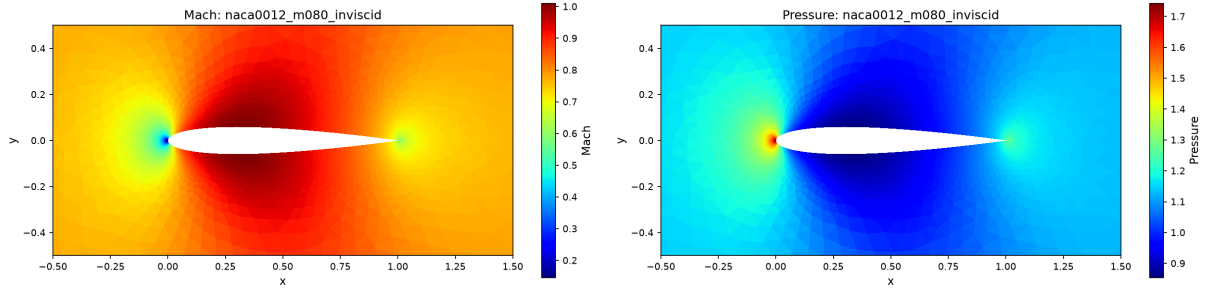


Figure 4: Mach and pressure contours (near-body): NACA 0012, $M = 0.8$, inviscid.

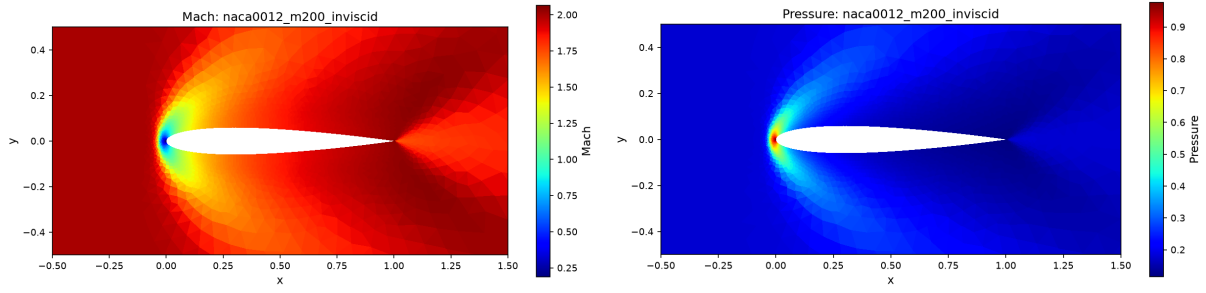


Figure 5: Mach and pressure contours (near-body): NACA 0012, $M = 2.0$, inviscid.

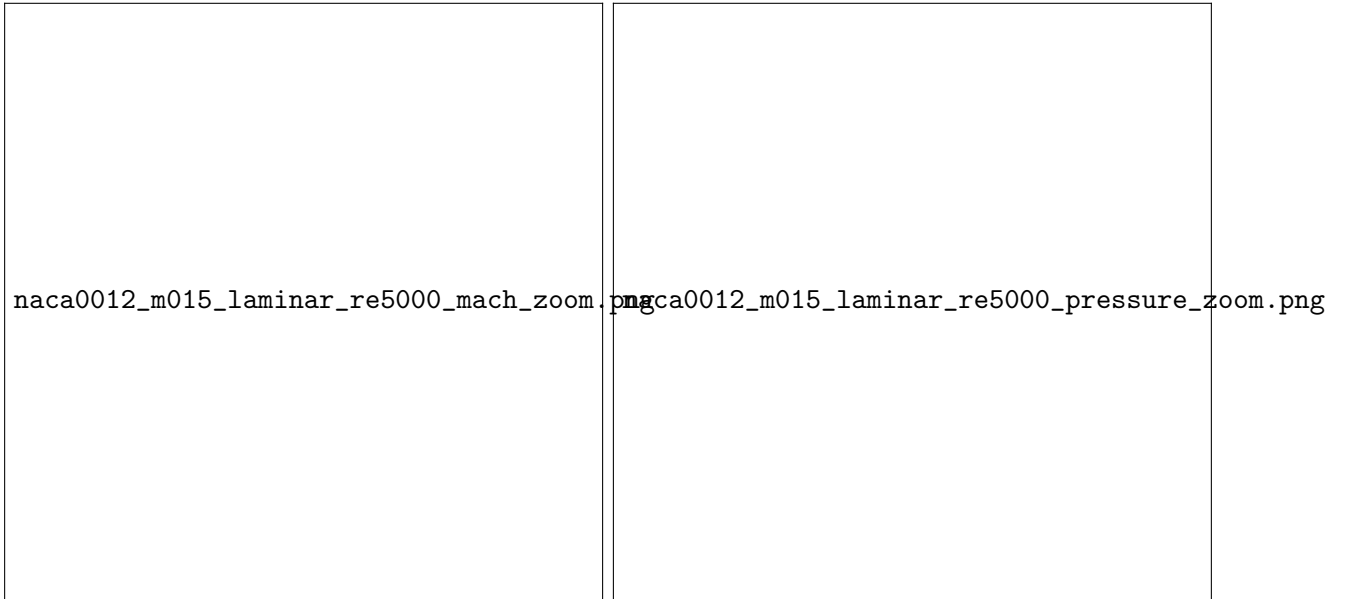


Figure 6: Mach and pressure contours (near-body): NACA 0012, $M = 0.15$, $Re\ 5000$.

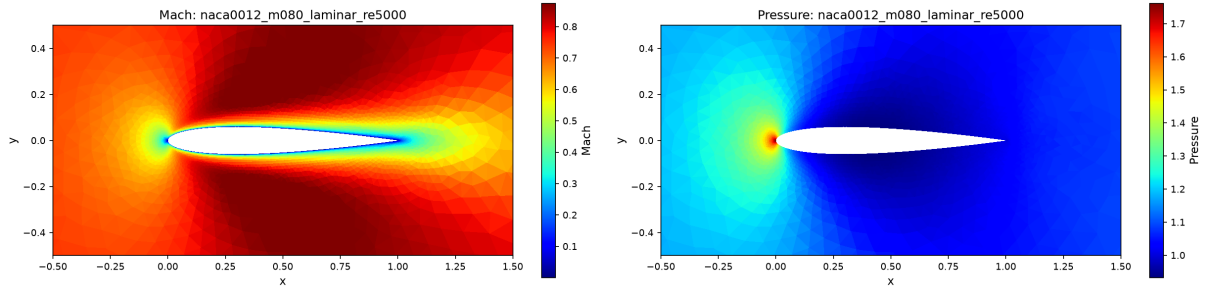


Figure 7: Mach and pressure contours (near-body): NACA 0012, $M = 0.8$, $\text{Re } 5000$.

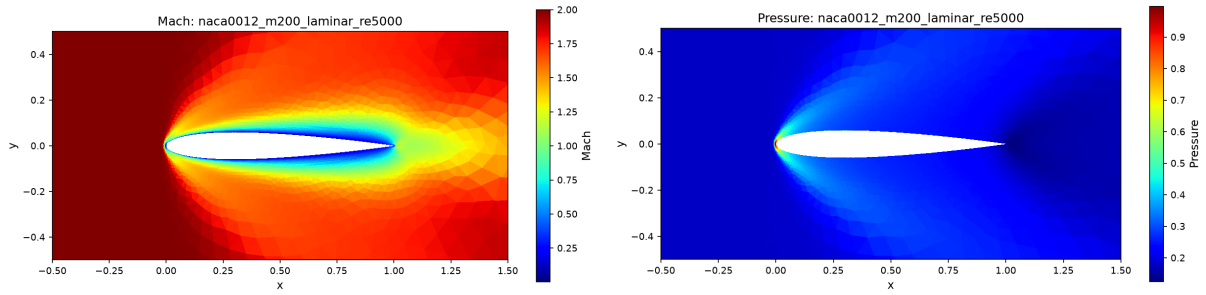


Figure 8: Mach and pressure contours (near-body): NACA 0012, $M = 2.0$, $\text{Re } 5000$.

9.3 Cylinder Reynolds 20

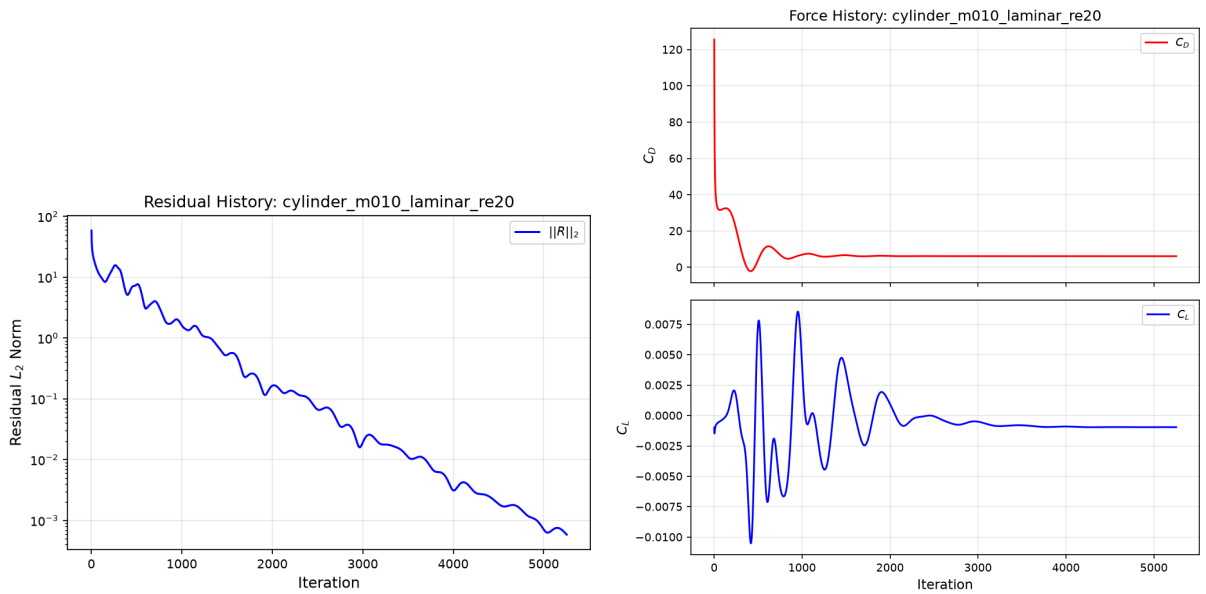


Figure 9: Residual and force history for cylinder, $\text{Re } 20$.

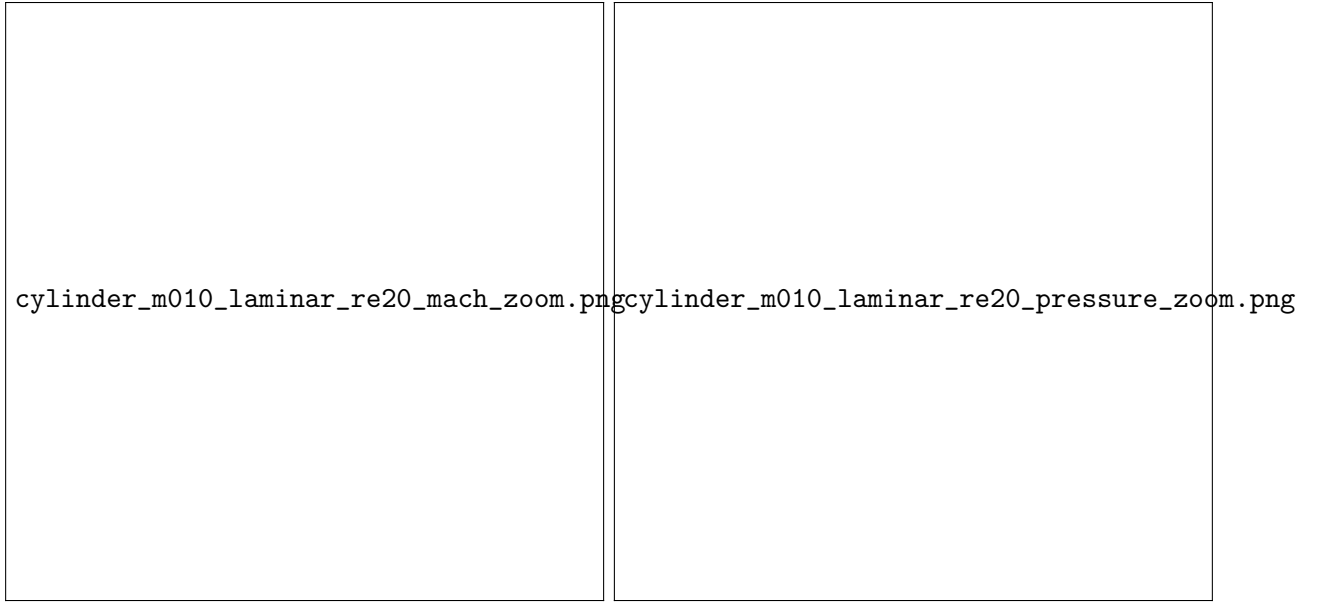


Figure 10: Mach and pressure contours (near-body): cylinder, Re 20.

9.4 Cylinder Reynolds 200 Vortex Street

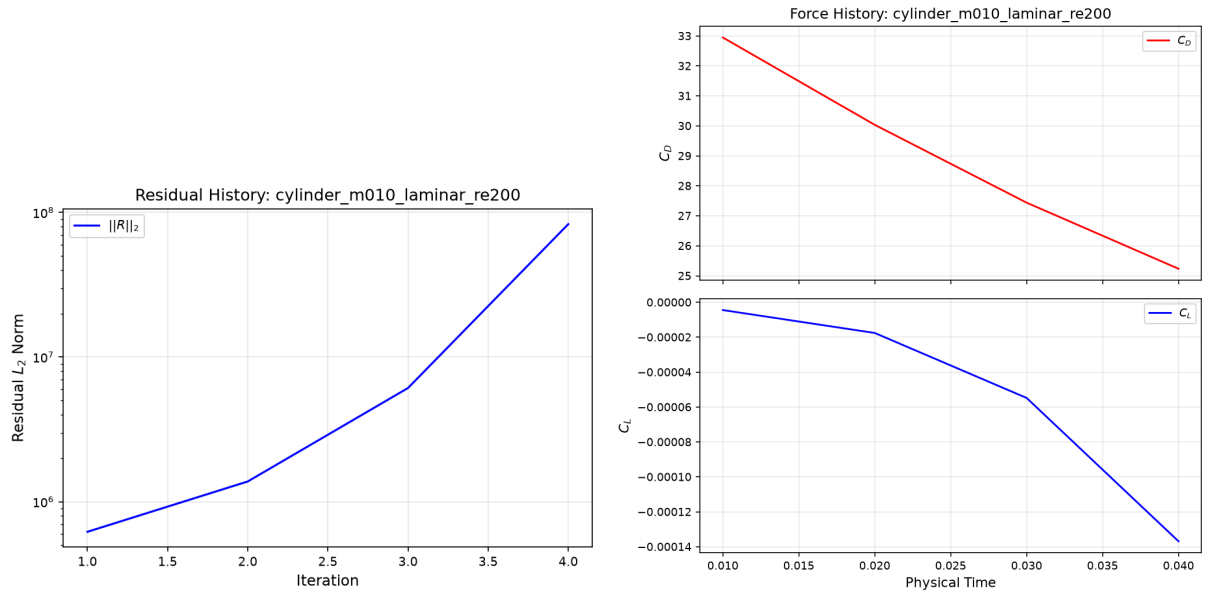
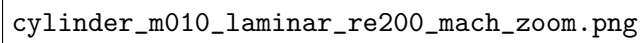


Figure 11: Residual and force history for cylinder, Re 200.

The image is a Mach contour plot showing the wake region of a cylinder at a Reynolds number of 200. The plot displays a series of alternating high and low Mach number contours, indicating the presence of a vortex street. The contours are concentrated in the wake region, showing the periodic shedding of vortices. The overall shape of the wake is elongated and symmetric about the horizontal axis.

cylinder_m010_laminar_re200_mach_zoom.png

Figure 12: Mach contour (wake region): cylinder, Re 200 vortex street.

The figure is a velocity-x contour plot showing the wake region of a cylinder at a Reynolds number of 200. The plot displays a series of alternating high and low velocity regions, characteristic of a vortex street. The contours are labeled with numerical values, and the overall shape of the wake is elongated and periodic. The plot is titled 'cylinder_m010_laminar_re200_velocityx_zoom.png'.

cylinder_m010_laminar_re200_velocityx_zoom.png

Figure 13: Velocity- x contour (wake region): cylinder, Re 200 vortex street.

10 Parallel Validation

Rank-count comparison tests were run for NACA 0012 $M = 0.15$ inviscid and cylinder Re 20 at $n_p = 4$ and $n_p = 8$.

11 Limitations and Failure Analysis

- The Barth–Jespersen limiter provides minimum acceptable oscillation control; Venkatakrishnan or MLP limiting would improve smooth-region accuracy.

- The LU-SGS implicit solver uses a simplified scalar spectral-radius Jacobian approximation. A block or matrix-free approach would improve convergence rates.
- The viscous gradient correction uses a simple direct correction along the cell-to-cell direction; a diamond-path or iterative correction would be more accurate on highly skewed meshes.